

Mahsa Amani

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Research Interests

- 📖 Large Language Models
- 📖 Natural Language Processing
- 📖 Deep Learning

Education

Max Planck Institute for Software Systems (MPI-SWS)	Saarbrücken, Germany
PH.D. IN COMPUTER SCIENCE	Jun. 2024 - Present
• Supervised by Prof. Krishna P. Gummadi	
Sharif University of Technology (SUT)	Tehran, Iran
B.SC. IN COMPUTER ENGINEERING	Sep. 2018 - Feb. 2023
• GPA: 18.03/20	
Farzanegan High School (National Organization for Development of Exceptional Talents (NODET))	Tabriz, Iran
DIPLOMA IN MATHEMATICS AND PHYSICS	Sep. 2011 - Jun. 2018
• GPA: 19.97/20	

Academic Honors

2022	Gifted Student Award Entrance exam waiver for graduate program, Sharif University of Technology	Iran
2018-2023	Membership In Iran's National Elites Foundation	Iran
2018	Rank 51st In Mathematics and Physics B.Sc. entrance exam among more than 144,000 participants	Iran
2017	Accepted In the First Exam of Mathematics Olympiad	Iran

Relevant Courses

Data Science	4/4
Natural Language Processing	20/20
Artificial Intelligence	20/20
Advanced Information Retrieval	19.3/20
Probability and Statistics	20/20
Data Structure and Algorithm	18.4/20

Languages

English	IELTS Overall Bandscore: 7.5 (🔊Listening: 9, 📖Reading: 7, 🗣Speaking: 6.5, ✍Writing: 6.5)
German	Elementary
Persian	Native Language
Azerbaijani	Mother Tongue

Publications

In Agents We Trust, but Who Do Agents Trust? Latent Source Preferences Steer LLM Generations	
M. A. KHAN, M. AMANI, S. DAS, B. GHOSH, Q. WU, K. P. GUMMADI, M. GUPTA, A. RAVICHANDER	Jul. 2025
• ICML 2025 Workshop on Reliable and Responsible Foundation Models	
Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs	
Q. WU, S. DAS, M. AMANI, B. GHOSH, M. A. KHAN, K. P. GUMMADI, M. B. ZAFAR	Jul. 2025
• ICML 2025 Workshop on The Impact of Memorization on Trustworthy Foundation Models	
Revisiting Privacy, Utility, and Efficiency Trade-offs when Fine-Tuning Large Language Models	
S. DAS, C. KOLLING, M. A. KHAN, M. AMANI, B. GHOSH, Q. WU, T. SPEICHER, K. P. GUMMADI	Feb. 2025
• ArXiv	
Linguistic Resources and Transformer-based Models for the Machine Translations between Luri and Yazdi Dialects versus Standard Persian	
M. AMANI, M. NOURI, Z. BAHMANI, M. MIRBEYGI, N. HASHEMI, E. ASGARI, M. SOLEYMANI BAGHSHAH, H. BEIGY, A. MOVAGHAR	Feb. 2024
• Journal of the Linguistics Society of Iran	

Skills

Programming

Python, R, C/C++, JAVA, SQL, $\LaTeX$

ML tools

Huggingface, VLLM, Pytorch, Tensorflow, Keras, Scikit-learn, Numpy, Pandas

Research Experience

Multigenerator, Multidomain, and Multilingual Black-Box Machine-Generated Text Detection

SEM EVAL2024

• Development of automatic systems to identify machine-generated text with the goal of mitigating its potential misuse

Pipelines for low-resource Iranian Languages

RESEARCH ASSISTANT

• Developing pipelines for low-resource languages such as Azerbaijani, Luri, Yazdi, and Kurdi

A Large Language Model for Persian

RESEARCH ASSISTANT

• Development of a multilingual large language model fine-tuned for Persian

Explainable Detection of Online Sexism

SEM EVAL2023

• Development of English-language models for sexism detection with fine-grained classifications for sexist content from Gab and Reddit

Iran

Dec. 2023 - Jan. 2024

Iran

Sep. 2022 - Dec. 2023

Iran

Oct. 2023

Iran

Feb. 2023

Work Experience

Teaching Assistant

MAX PLANCK INSTITUTE

• Efficient Training of Large Language Models: From Basics to Fine-Tuning , Prof. Krishna Gummadi

Junior Data Scientist

ZAREBIN

• Developing a Shopping Information Retrieval System using Deep Learning Methods

Teaching Assistant

SHARIF UNIVERSITY OF TECHNOLOGY (SUT)

• Natural Language Processing, Dr. Ehsaneddin Asgari

• Probability and Statistics, Dr. Ali Sharifi

• Database Design, Dr. Abbas Heydarnoori

• Computer Architecture, Dr. Hossein Asadi

• Python Programming, Dr. Issa Moradnejad

• Digital System Design, Dr. Farshad Baharvand

• Digital Design, Dr. Laleh Arshadi

Junior Data Analyst

VIVETENNIS

• Developing and Analysing Robot Monitoring System

Undergraduate Student Internship

AIMEDIC

• Eye Disease Diagnosis using Artificial Intelligence

Saarland, Germany

Oct. 2025 - Feb. 2026

Tehran, Iran

Oct. 2022 - Apr. 2023

Tehran, Iran

Sep. 2020 - Jul. 2023

Tehran, Iran

Jun. 2022 - Aug. 2022

Tehran, Iran

Jul. 2021 - Sep. 2021

Attended Conferences and Workshops

2025 Association for Computational Linguistics, Vienna, Austria

2025 ACM International Conference on Web Search and Data Mining, Hannover, Germany

July 2025

March, 2025

Projects

Unveiling Hospital Cost Drivers

EXPLAINABLE AND AGENTIC AI APPROACH FOR UNVEILING HOSPITAL COST DRIVERS

Germany

Mar. 2025 - August. 2025

Iranian Azerbaijani NLP Pipeline

THE LANGUAGE MODEL, RESOURCES, AND COMPUTATIONAL PIPELINES FOR THE UNDER-RESOURCED IRANIAN AZERBAIJANI

Iran

Jan. 2023 - May. 2023

3D Vision Transformers and Convolutional Models for Segmentation Tasks

BACHELOR'S THESIS ON 3D MRI BRAIN TUMOR SEGMENTATION

Iran

Jan. 2022 - Jan. 2023

Azerbaijani to Farsi Translation

DEVELOPING JOEYNMT FRAMEWORK FOR TRANSLATING AZERBAIJANI TO FARSI USING TRANSFORMERS AND SEQ2SEQ MODELS

Iran

Jun. 2022

Persian NER and Spell Corrector

FINE TUNING BERT MODEL FOR NAME ENTITY RECOGNITION AND LANGUAGE SPELL CORRECTION

Iran

May. 2022 - Jun. 2022

Persian QA Extractor

EXTRACTING QUESTIONS AND RESPECTIVE ANSWERS FROM A TEXT USING REGULAR EXPRESSIONS

Iran

April 2022

Turkish Preprocessing Pipeline

DEVELOPING A PREPROCESSING PIPELINE FOR TURKISH LANGUAGE

Iran

April 2022

Modern Information Retrieval

A PROJECT INCLUDING INDEXING, COMPRESSING, QUERY CORRECTION, LANGUAGE MODELS, CLASSIFIERS, AND CLUSTERING

Iran

Feb. 2021

TeleNurse

A LOGISTIC PROJECT FOR PROVIDING HEALTHCARE AND NURSING IMPLEMENTED USING PYTHON DJANGO, POSTGRESQL, AND DOCKER

Iran

Nov. 2022

Eye Disease Recognition

RECOGNITION OF 7 EYE DISEASES USING TRANSFER LEARNING IMPLEMENTED AS MY INTERNSHIP PROJECT

Iran

Sep. 2021